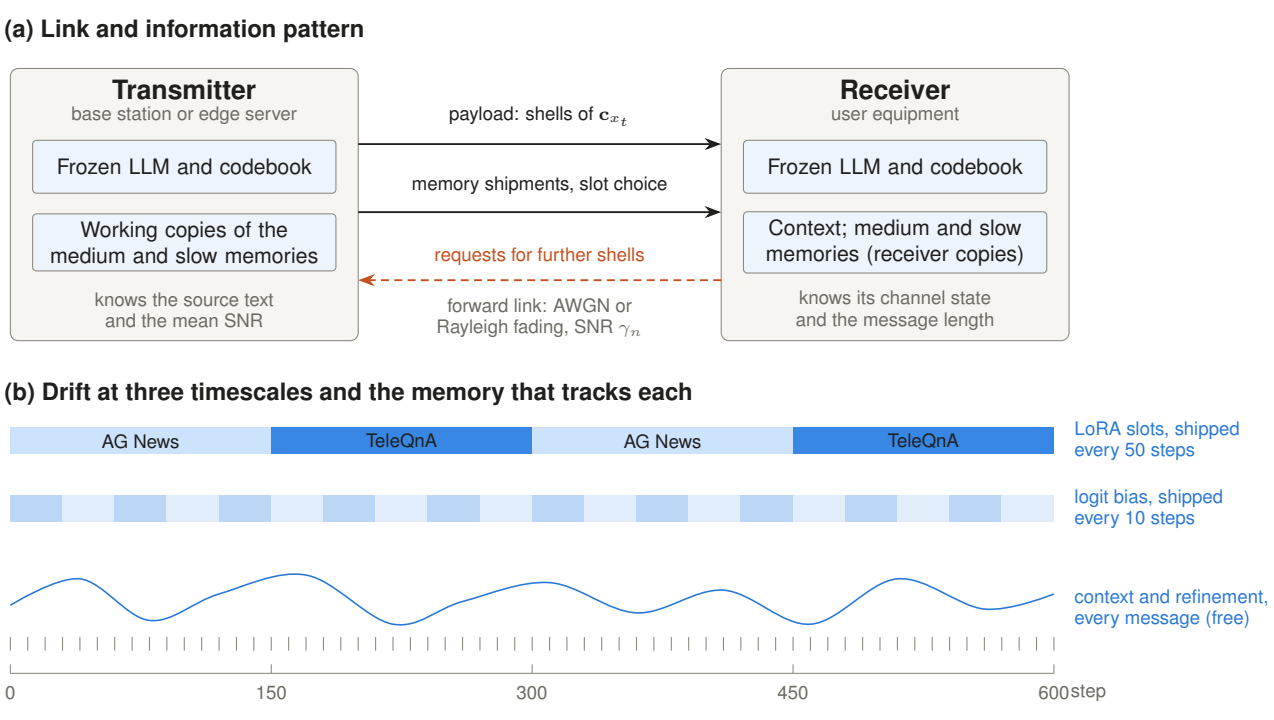

(a) Link and information pattern
Transmitter
base station or edge server
Frozen LLM and codebook
Working copies of the medium and slow memories
knows the source text and the mean SNR
payload: shells of $\mathbf{c}_{x_t}$
memory shipments, slot choice
requests for further shells
forward link: AWGN or Rayleigh fading, SNR γ_n
Receiver
user equipment
Frozen LLM and codebook
Context; medium and slow memories (receiver copies)
knows its channel state and the message length
(b) Drift at three timescales and the memory that tracks each
slow: domain ($\tau_{\text{slow}} = 150$)
AG News
TeleQnA
AG News
TeleQnA
LoRA slots, shipped every 50 steps
medium: topic ($\tau_{\text{med}} = 30$)
logit bias, shipped every 10 steps
fast: SNR γ_n, messages
context and refinement, every message (free)
0
150
300
450
600
step